

Page 1: Introduction to Java

Java Programming

Java is a high-level, object-oriented programming language developed by Sun Microsystems.

Features of Java:

- Platform Independent
- Object-Oriented
- Secure
- Robust
- Multithreaded
- Portable

Java code is compiled into bytecode, which runs on the Java Virtual Machine (JVM).

Page 2: OOP Concepts

Object-Oriented Programming

Java follows Object-Oriented Programming principles.

Class

A class is a blueprint for creating objects.

Object

An object is an instance of a class.

Encapsulation

Wrapping data and methods together.

Inheritance

One class acquires properties of another class.

Polymorphism

One interface, many implementations.

Abstraction

Hiding implementation details from users.

Page 3: Java Collections

Collections Framework

Java Collections Framework provides classes to store and manipulate data.

Common Collections:

- ArrayList
- LinkedList
- HashSet
- TreeSet
- HashMap
- PriorityQueue

HashMap

Stores data as key-value pairs.

Allows fast insertion and retrieval.

Does not maintain insertion order.

Page 4: Exception Handling

Exception Handling

Exception handling prevents program crashes.

Keywords:

- try
- catch
- finally
- throw
- throws

Example exceptions:

- ArithmeticException
- NullPointerException
- IOException

Page 5: Multithreading

Multithreading

Multithreading allows multiple tasks to execute simultaneously.

Advantages:

- Better CPU utilization
- Faster execution
- Improved responsiveness

Thread Lifecycle:

- New
- Runnable
- Running
- Waiting
- Terminated

Page 6: JDBC

JDBC

JDBC stands for Java Database Connectivity.

Steps:

1. Load Driver
2. Create Connection
3. Create Statement
4. Execute Query
5. Process Result
6. Close Connection

JDBC is commonly used to connect Java applications with MySQL and PostgreSQL databases.

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